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Trincomalee Campus, Eastern University, Sri Lanka
Faculty of Applied Science
Department of Computer Science
Bachelor of Computer Science (2016/2017)
Year I Semester II Examination (February 2019)
CO1225 - Computer Architecture

Answer All Questions **Time: 02 Hours**

Q1. 1) Digital electronics only understand two states, ON and OFF. This is why digital electronics use the binary number system.

a. Convert the following hexadecimal numbers to binary numbers:

i. $FD73_{16}$;

ii. 75.53_{16}. (2x2=4 marks)

b. Convert 34.625 base 10 to binary, base 3, octal, and hexadecimal. Any fractions that do not terminate should be truncated to 4 digits in the fractional part. (2x4=8 marks)

2) Signed numbers' are required to encode negative numbers in binary number system.

a. What are the three common ways of representing signed numbers? (2 marks)

b. Let the decimal numbers A=54, B= -77, give their 8-bit 2's complement representation. Compute A-B in 2's complement, is it overflow? Why or why not? (4 marks)

c. Give the 16-bit 2's complement representation of A, and B. Compute A-B. Is it overflow? (2 marks)

3) Write down whether the following statements are True or False.

a. Registers are inside the main memory on computer mother board

b. Von Neumann architecture has data and instructions in the same memory space.

c. Dynamic RAM is faster than Static RAM in access time.

d. In modern computers, CPU and Memory are connected by BUS

e. The density of data on the hard disk does not affect data access speed.

(1x5=5 marks)

[Total 25 Marks]

Q2. 1) For the following truth table, where x is don't care state

A	B	C	D	F
0	0	0	0	1
0	0	0	1	X
0	0	1	0	1
0	0	1	1	0
0	1	0	0	X
0	1	0	1	X
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	X
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

a. Draw the Karnaugh map for F (4 marks)

b. Identify all the sub-cubes (max size, min number) on another copy of your K-map. (4 marks)

c. Use the sub-cubes identified in part b. to produce a minimum 'sum of products' expression for F. (3 marks)

d. Draw a logic the logic circuit for F (4 marks)

2) A full adder receives three bits (a, b, cin) as input and output the sum bit and carry bit (s, cout).

a. Write the truth table of the full adder and the corresponding canonical product-of maxterms expressions. (5 marks)

b. Using basic gates draw a logic circuit for the full adder with. (5 marks)

[Total 25 Marks]

Q3. 1) A computer science graduate should have clear understanding about computer architecture and organization.
a) What are the major structural components of CPU? Briefly describe.
b) Describe the functional view of a computer with the aid of a diagram.
c) Discuss the Memory Hierarchy of a computer.
d) How the computer architecture differs from Computer organization? Briefly explain.
(3x4=12 marks)

2) Briefly describe the of Flynn's taxonomy for classifying computer architectures. (3 marks)

3) Briefly describe the following terms:
a) MIPS;
b) MFLOPS;
c) Word length. (1x3=3 marks)

4) What is meant by instruction set architecture of a computer? (3 marks)

5) List three main addressing modes that are used on various platforms and architectures and describe them briefly. (4 marks)

[Total 25 Marks]

Q4. 1) Pipelining is a technique used in processors.
a. Describe the purpose of pipelining using suitable examples?
b. State three different types of pipeline hazards. Briefly explain any two of them.
c. Distinguish between RISC and CISC
d. Compare the Superscalar Architecture with simple pipelined architecture.
(3x4=12 marks)

2) Parallel processing is a method to improve computer system performance by executing two or more instructions simultaneously.
a. What are the goals of parallel processing?
b. What is Amdahl's Law?
c. Compute, if 90% of a calculation can be parallelized (i.e. 10% is sequential) then the maximum speedup, which can be achieved on 5 processors. (3x3=9 marks)

3) Briefly explain the followings:
a. Shared memory;
b. Distributed Shared memory. (2x2=4 marks)

[Total 25 Marks]